MATRIX · INVERSE

# When structure makes the inverse cheap

Each of these is invertible under a condition readable off the matrix, and each has an inverse that inherits the same structure. Recognising the type is worth more than any general algorithm.

6
TYPES

## THE INVERSE IS FREE — 1

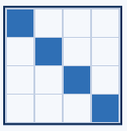


### Orthogonal | $Q^{\mathsf{T}}Q = I$

| INVERTIBLE WHEN | INVERSE | COST |
|---|---|---|
| always — $\det Q = \pm 1$ | $Q^{-1} = Q^{\mathsf{T}}$ | none — transpose |

The only row where no arithmetic happens at all. Always invertible, and the inverse is a relabelling of the entries — which is why orthogonal factors are worth arranging for in a factorization.

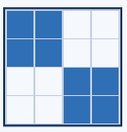
## THE INVERSE IS ENTRYWISE OR BLOCKWISE — 2



### Diagonal | $D = \operatorname{diag}(d_1, \ldots, d_n)$

| INVERTIBLE WHEN | INVERSE | COST |
|---|---|---|
| every $d_i \neq 0$ | $\operatorname{diag}(1/d_1, \ldots, 1/d_n)$ | $n$ divisions |

Reciprocate the diagonal. The condition is visible without computing anything, since the determinant is the product of exactly those entries.

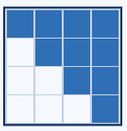


### Block diagonal | $\operatorname{diag}(A_1, \ldots, A_k)$

| INVERTIBLE WHEN | INVERSE | COST |
|---|---|---|
| every block $A_i$ invertible | $\operatorname{diag}(A_1^{-1}, \ldots, A_k^{-1})$ | invert each block |

The diagonal case with numbers replaced by blocks. An $n \times n$ matrix in $k$ equal blocks costs $k$ inversions of size $n/k$ rather than one of size $n$ — a saving of $k^2$, since the cost is cubic.

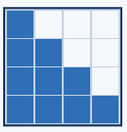
## THE INVERSE COMES FROM SUBSTITUTION — 2



### Upper triangular | zeros below the diagonal

| INVERTIBLE WHEN | INVERSE | COST |
|---|---|---|
| every diagonal entry $\neq 0$ | upper triangular | back substitution, $n^2$ |

Solve for the columns from the bottom up. The inverse stays upper triangular, which is why an LU factorization is worth keeping rather than discarding after the first solve.

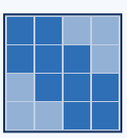


### Lower triangular | zeros above the diagonal

| INVERTIBLE WHEN | INVERSE | COST |
|---|---|---|
| every diagonal entry $\neq 0$ | lower triangular | forward substitution, $n^2$ |

The mirror image, worked from the top down. Together with the row above, these are the two solves an LU-based inverse actually performs.

## STRUCTURE PRESERVED BUT NO SHORTCUT — 1



### Symmetric | $A = A^{\mathsf{T}}$

| INVERTIBLE WHEN | INVERSE | COST |
|---|---|---|
| $\det(A) \neq 0$ | also symmetric | half of a general solve |

The one row whose condition is not local — symmetry constrains the inverse without deciding whether it exists. The saving is real but modest: only one triangle need be computed. When the matrix is also positive definite, Cholesky halves it again.

Two things recur down the table. The invertibility condition is always local — a diagonal entry, a block, never a determinant — and the inverse always belongs to the same family, which is what makes these routes composable. General row reduction discards both advantages and should be the fallback rather than the first move.